

Bayes Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Given $P(B) \neq 0$

$P(A|B)$ = Posterior

$P(B|A)$ = likelihood

$P(A)$ = Prior

$P(B)$ = Evidence

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{--- (1)}$$

$$A \cap B = B \cap A$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A \cap B)}{P(A)}$$

$$P(A \cap B) = P(B|A) * P(A) \text{ --- (1)}$$

$$\therefore P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Bayes Theorem